

Design of an Arduino-Based LED Stack Game with Hybrid Ultrasonic and Push-Button Control for Enhanced Human–Machine Interaction

Abstract

This project presents the design and implementation of an interactive LED stack game using an Arduino UNO and a 4-in-1 MAX7219 dot matrix display. The proposed system introduces a hybrid dual-input control mechanism, where an ultrasonic sensor enables contactless horizontal positioning of the moving LED stack, and a push button is used to lock the falling block into place. The objective is to enhance accessibility, intuitiveness, and user engagement compared to conventional button-only embedded games. The system demonstrates principles of embedded system design, real-time control, and human–machine interaction, making it suitable for educational and assistive applications.

1. The Problem

Embedded LED-based games developed using microcontrollers typically rely on **single-input mechanisms**, most commonly push buttons or switches. While these interfaces are simple and reliable, they suffer from several limitations:

- Restricted interactivity due to discrete input states
- Lack of intuitive or natural control mechanisms
- Reduced accessibility for users with motor impairments or limited dexterity
- Minimal consideration of human–machine interaction principles

Existing LED stack games often prioritize functional implementation over user experience, resulting in systems that are technically correct but interactionally limited. There is a clear need for embedded gaming systems that offer **more intuitive, inclusive, and flexible input methods** without significantly increasing system complexity or cost.

2. The Solution

The proposed solution introduces a **dual-input hybrid control system** for an Arduino-based LED stack game:

- An **ultrasonic sensor** is used to control the **horizontal movement** of the LED stack in real time based on hand distance, enabling contactless and continuous control.
- A **push button** is used to **lock or stop the falling block**, providing precise and deliberate confirmation of the stacking action.

By separating **continuous motion control** (ultrasonic sensor) from **discrete action control** (push button), the system combines the advantages of both sensor-based and tactile interfaces. This design improves intuitiveness, reduces physical effort, and enhances overall interaction quality compared to conventional button-only systems.

3. Benefits and Strategic Alignment

Key Benefits

- **Improved Accessibility:** Contactless control reduces dependency on fine motor skills.
- **Enhanced User Experience:** Natural hand-based interaction increases engagement.
- **Higher Educational Value:** Demonstrates hybrid input systems and HMI concepts.
- **Low-Cost Innovation:** Uses commonly available sensors and components.
- **Modular Design:** Easily extendable for future enhancements.

Strategic Alignment

The project aligns with broader objectives of:

- Human-centered embedded system design
- Inclusive and assistive technology development
- Skill-based learning in electronics and microcontrollers
- Innovation-driven academic project evaluation

4. Implementation Details

4.1 Architecture / Design

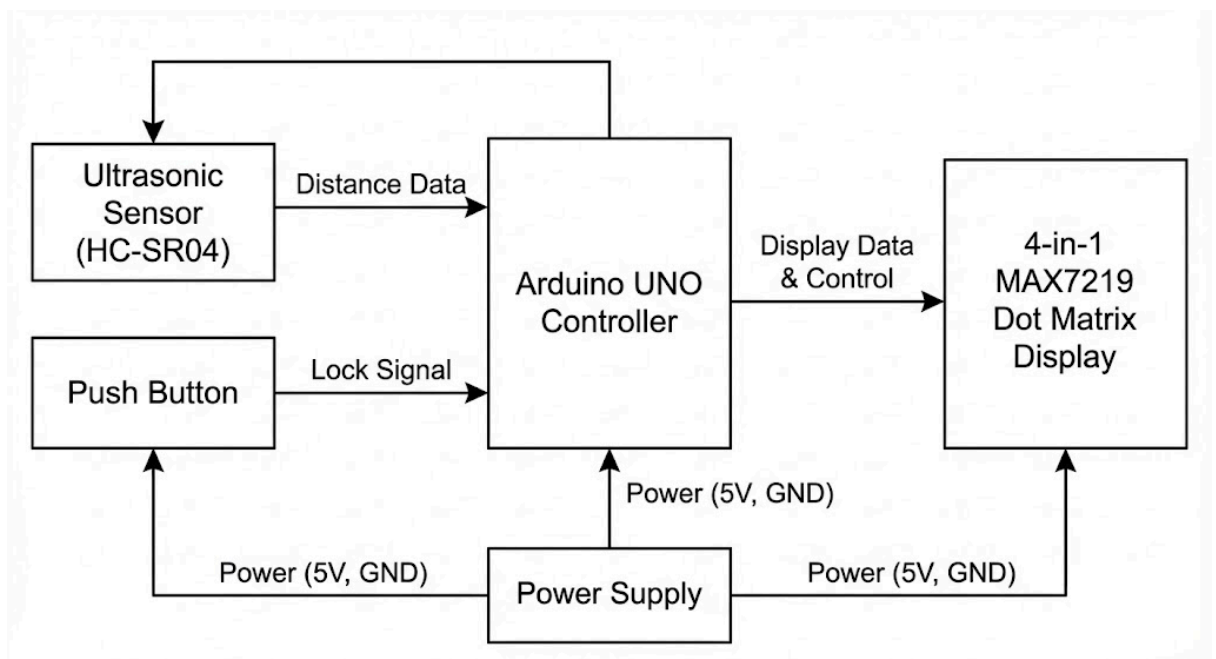
- **Controller:** Arduino UNO executes game logic and input processing.
- **Display Unit:** 4-in-1 MAX7219 dot matrix displays the moving and stacked LEDs.
- **Input Layer:**
 - Ultrasonic sensor provides continuous distance data mapped to horizontal position.
 - Push button provides discrete confirmation to lock the stack.

Data Flow:

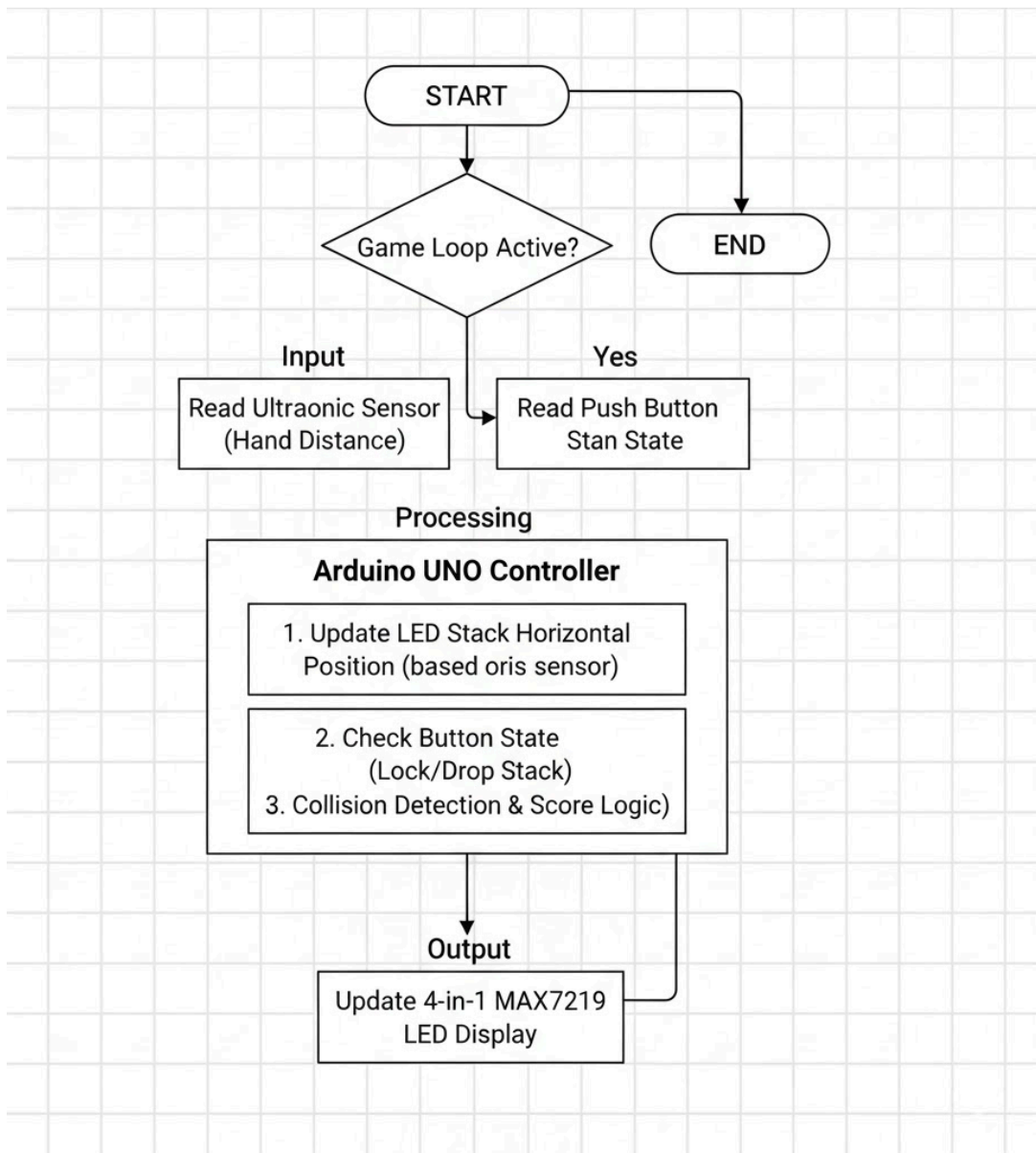
Sensor inputs → Arduino processing → LED matrix visual output

4.2 Visuals

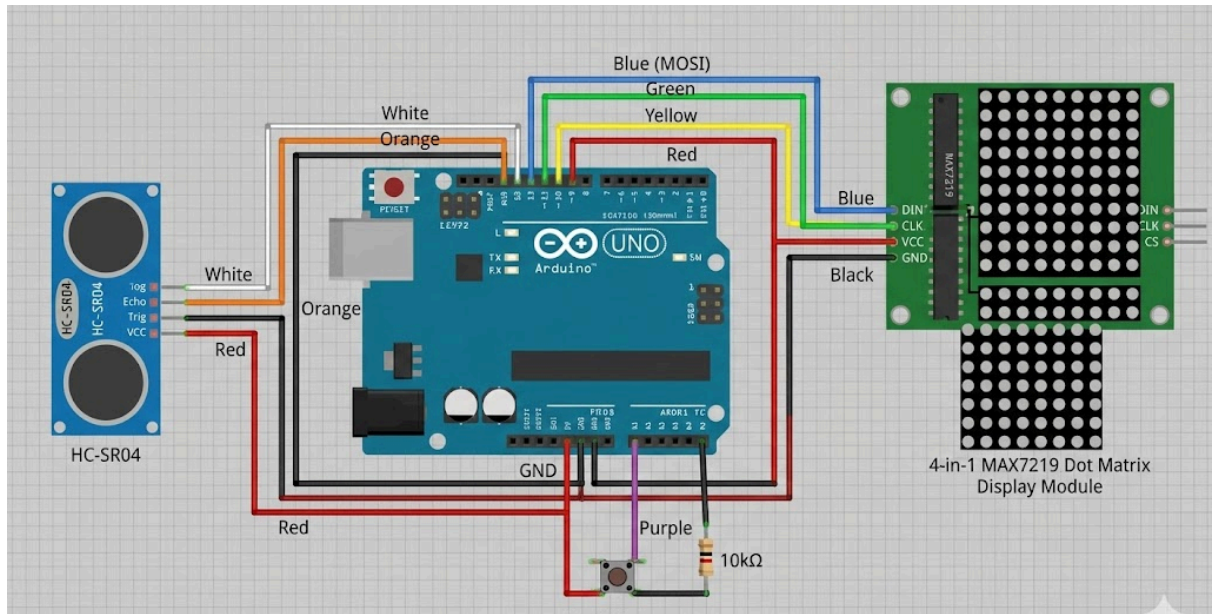
- System block diagram



- Input–processing–output flowchart



- LED matrix stacking sequence diagram



4.3 Materials / Tools

- Arduino UNO
- 4-in-1 MAX7219 dot matrix display
- Ultrasonic sensor (HC-SR04 or equivalent)
- Push button
- Arduino IDE (Embedded C)
- Jumper wires and power supply

5. Results / Conclusion

The proposed system demonstrates that integrating sensor-based and manual inputs significantly improves interactivity and accessibility in embedded LED games. The hybrid control approach allows intuitive positioning while retaining precise control over critical actions. This design successfully addresses limitations of conventional single-input systems and establishes a foundation for more inclusive and human-centered embedded applications.